AUM – U07079548 – HW4 – CLOUD COMPUTING **(The questions are not ordered sequentially due to order of execution)**

VM is running

```
Aum@hw4-server-vm:~$ Aum@hw4-server-vm:~$ sudo systemctl status service1
● service1.service - HW4 Service 1 - Python HTTP Web Server
     Loaded: loaded (/etc/systemd/system/service1.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-05 18:21:34 UTC; 8min ago
   Main PID: 2943 (python)
      Tasks: 8 (limit: 1136)
     Memory: 51.3M
        CPU: 6.441s
     CGroup: /system.slice/service1.service
             └─2943 /opt/service1/venv/bin/python /opt/service1/service1.py

Mar 05 18:21:34 hw4-server-vm systemd[1]: Started service1.service - HW4 Service 1 - Python HTTP Web Server.
Mar 05 18:21:40 hw4-server-vm python[2943]: [Service 1] Listening on 0.0.0.0:80
Mar 05 18:21:40 hw4-server-vm python[2943]: [Service 1] Bucket: gs://aum-hw2-u07079548/hw2/^C
```

Question 1

```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -v http://136.113.118.31/0.html
*   Trying 136.113.118.31:80...
* Connected to 136.113.118.31 (136.113.118.31) port 80
* using HTTP/1.x
> GET /0.html HTTP/1.1
> Host: 136.113.118.31
> User-Agent: curl/8.10.1
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 05 Mar 2026 18:51:48 GMT
< Content-Type: text/html; charset=utf-8
< Content-Length: 99334
<
<!DOCTYPE html>
<html>
<body>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse
cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
<p>
<a HREF="13781.html"> This is a link </a>
<p>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse
cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
<p>
<a HREF="1326.html"> This is a link </a>
```

Question 2(Also Question 5)

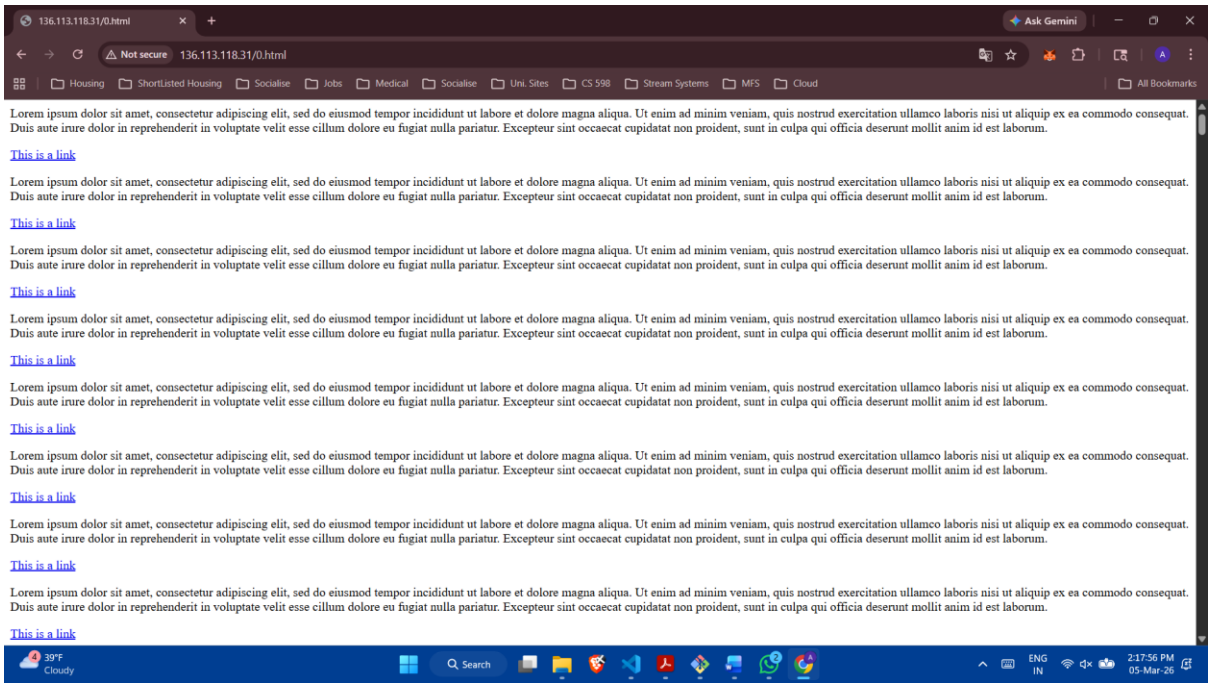
```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -v http://136.113.118.31/filedoesnotexist.html
*   Trying 136.113.118.31:80...
* Connected to 136.113.118.31 (136.113.118.31) port 80
* using HTTP/1.x
> GET /filedoesnotexist.html HTTP/1.1
> Host: 136.113.118.31
> User-Agent: curl/8.10.1
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 404 Not Found
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 05 Mar 2026 18:53:22 GMT
< Content-Length: 13
<
404 Not Found* shutting down connection #0
```

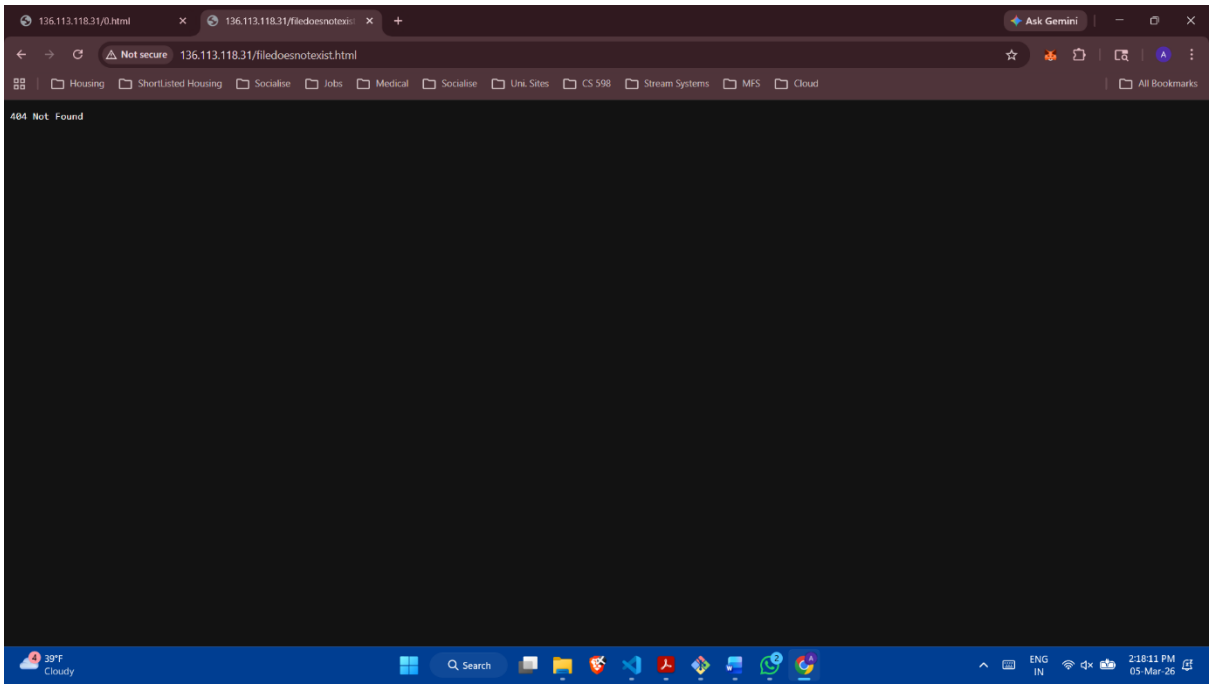
Question 3 (Also Question 5)

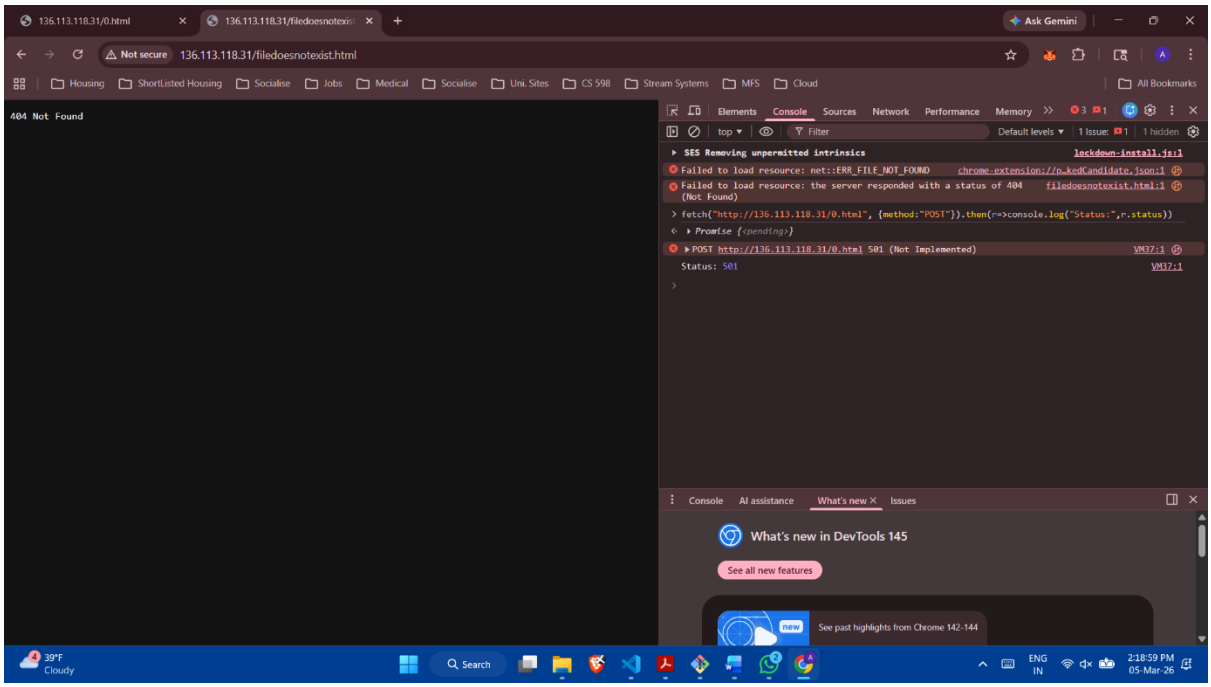
```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -v -X POST http://136.113.118.31/0.html
*   Trying 136.113.118.31:80...
* Connected to 136.113.118.31 (136.113.118.31) port 80
* using HTTP/1.x
> POST /0.html HTTP/1.1
> Host: 136.113.118.31
> User-Agent: curl/8.10.1
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 501 Not Implemented
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 05 Mar 2026 18:55:49 GMT
< Content-Length: 19
<
501 Not Implemented* shutting down connection #0
```

```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -v -H "X-country: Iran" http://136.113.118.31/0.html
*   Trying 136.113.118.31:80...
* Connected to 136.113.118.31 (136.113.118.31) port 80
* using HTTP/1.x
> GET /0.html HTTP/1.1
> Host: 136.113.118.31
> User-Agent: curl/8.10.1
> Accept: */*
> X-country: Iran
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 403 Forbidden
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 05 Mar 2026 19:27:32 GMT
< Content-Length: 37
<
Forbidden: export restrictions apply.* shutting down connection #0

Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -H "X-country: Cuba" http://136.113.118.31/0.html
Forbidden: export restrictions apply.
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -H "X-country: North Korea" http://136.113.118.31/0.html
Forbidden: export restrictions apply.
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
$ curl -H "X-country: Syria" http://136.113.118.31/0.html
Forbidden: export restrictions apply.
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw4 (main)
```

Question 6







Question 7 - Logs


```
2026-03-05 14:17:45.926  hw4-server-vm  {"bucket":"aum-hw2-u07079548", "event":"FILE_NOT_FOUND", "file_path":"hw2/favicon.ico"}
Investigate log | Copy | Expand nested fields | Hide log summary
{
  insertId: "1sflb8mf90u5ml"
  jsonPayload: {
    bucket: "aum-hw2-u07079548"
    event: "FILE_NOT_FOUND"
    file_path: "hw2/favicon.ico"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:17:45.926202628Z"
  resource: {2}
  severity: "WARNING"
  timestamp: "2026-03-05T19:17:45.926202628Z"
}
```

```
2026-03-05 14:18:09.969  hw4-server-vm  {"bucket":"aum-hw2-u07079548", "event":"FILE_NOT_FOUND", "file_path":"hw2/filedoesnotexist.html"}
Investigate log | Copy | Expand nested fields | Hide log summary
{
  insertId: "1jjco00f9rjm4x"
  jsonPayload: {
    bucket: "aum-hw2-u07079548"
    event: "FILE_NOT_FOUND"
    file_path: "hw2/filedoesnotexist.html"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:18:09.969385229Z"
  resource: {2}
  severity: "WARNING"
  timestamp: "2026-03-05T19:18:09.969385229Z"
}
```

2026-03-05 14:18:56.027 hw4-server-vm {"event":"UNSUPPORTED_METHOD", "method":"POST", "path":"/0.html"}

Investigate log | Copy | Expand nested fields | Hide log summary


```
{
  insertId: "17ue3tmf4vkshm"
  jsonPayload: {
    event: "UNSUPPORTED_METHOD"
    method: "POST"
    path: "/0.html"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:18:56.027139167Z"
  resource: {2}
  severity: "WARNING"
  timestamp: "2026-03-05T19:18:56.027139167Z"
}
```


2026-03-05 14:27:32.765 hw4-server-vm {"country":"Iran", "event":"FORBIDDEN_COUNTRY", "path":"/0.html", "remote_addr":"128.197.28.143"}

Investigate log | Copy | Expand nested fields | Hide log summary


```
{
  insertId: "152g7fmeif9l0"
  jsonPayload: {
    country: "Iran"
    event: "FORBIDDEN_COUNTRY"
    path: "/0.html"
    remote_addr: "128.197.28.143"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:27:32.765479529Z"
  resource: {2}
  severity: "CRITICAL"
  timestamp: "2026-03-05T19:27:32.765479529Z"
}
```


2026-03-05 14:27:45.239 hw4-server-vm {"country":"Cuba", "event":"FORBIDDEN_COUNTRY", "path":"/0.html", "remote_addr":"128.197.28.143"}

Investigate log | Copy | Expand nested fields | Hide log summary


```
{
  insertId: "1iejz4af3wy8fi"
  jsonPayload: {
    country: "Cuba"
    event: "FORBIDDEN_COUNTRY"
    path: "/0.html"
    remote_addr: "128.197.28.143"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:27:45.239276531Z"
  resource: {2}
  severity: "CRITICAL"
  timestamp: "2026-03-05T19:27:45.239276531Z"
}
```


2026-03-05 14:28:00.295 hw4-server-vm {"country":"North Korea", "event":"FORBIDDEN_COUNTRY", "path":"/0.html", "remote_addr":"128.197.28.143"}

Investigate log | Copy | Expand nested fields | Hide log summary


```
{
  insertId: "ajluxjf4xz0as"
  jsonPayload: {
    country: "North Korea"
    event: "FORBIDDEN_COUNTRY"
    path: "/0.html"
    remote_addr: "128.197.28.143"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:28:00.295051053Z"
  resource: {2}
  severity: "CRITICAL"
  timestamp: "2026-03-05T19:28:00.295051053Z"
}
```

```
2026-03-05 14:28:11.918  hw4-server-vm  {"country":"Syria", "event":"FORBIDDEN_COUNTRY", "path":"/0.html", "remote_addr":"128.197.28.143"}
Investigate log   Copy   Expand nested fields   Hide log summary
{
  insertId: "1sdqafvf14gw3r"
  jsonPayload: {
    country: "Syria"
    event: "FORBIDDEN_COUNTRY"
    path: "/0.html"
    remote_addr: "128.197.28.143"
  }
  logName: "projects/u0709548-aum-hw1/logs/hw4-service1"
  receiveTimestamp: "2026-03-05T19:28:11.918966527Z"
  resource: {2}
  severity: "CRITICAL"
  timestamp: "2026-03-05T19:28:11.918966527Z"
}
```

Question 8 - SERVICE 2 – FORBIDDEN COUNTRY ALERT

```
Aum@hw4-service2-vm:~$ sudo journalctl -u service2 --no-pager | tail -20
Mar 05 18:22:03 hw4-service2-vm systemd[1]: Started service2.service - HW4 Servi
ce 2 - Forbidden Country Reporter.
Mar 05 18:22:08 hw4-service2-vm python[2848]: [Service 2] Listening on projects/
u0709548-aum-hw1/subscriptions/forbidden-requests-sub ...
Mar 05 18:57:44 hw4-service2-vm python[2848]: [2026-03-05T18:57:44.354046+00:00]
 FORBIDDEN REQUEST -- country=Iran, file=/0.html, remote_addr=128.197.28.143
Mar 05 19:27:33 hw4-service2-vm python[2848]: [2026-03-05T19:27:33.451848+00:00]
 FORBIDDEN REQUEST -- country=Iran, file=/0.html, remote_addr=128.197.28.143
Mar 05 19:27:45 hw4-service2-vm python[2848]: [2026-03-05T19:27:45.846318+00:00]
 FORBIDDEN REQUEST -- country=Cuba, file=/0.html, remote_addr=128.197.28.143
Mar 05 19:28:00 hw4-service2-vm python[2848]: [2026-03-05T19:28:00.949190+00:00]
 FORBIDDEN REQUEST -- country=North Korea, file=/0.html, remote_addr=128.197.28.
143
Mar 05 19:28:12 hw4-service2-vm python[2848]: [2026-03-05T19:28:12.582584+00:00]
 FORBIDDEN REQUEST -- country=Syria, file=/0.html, remote_addr=128.197.28.143
```

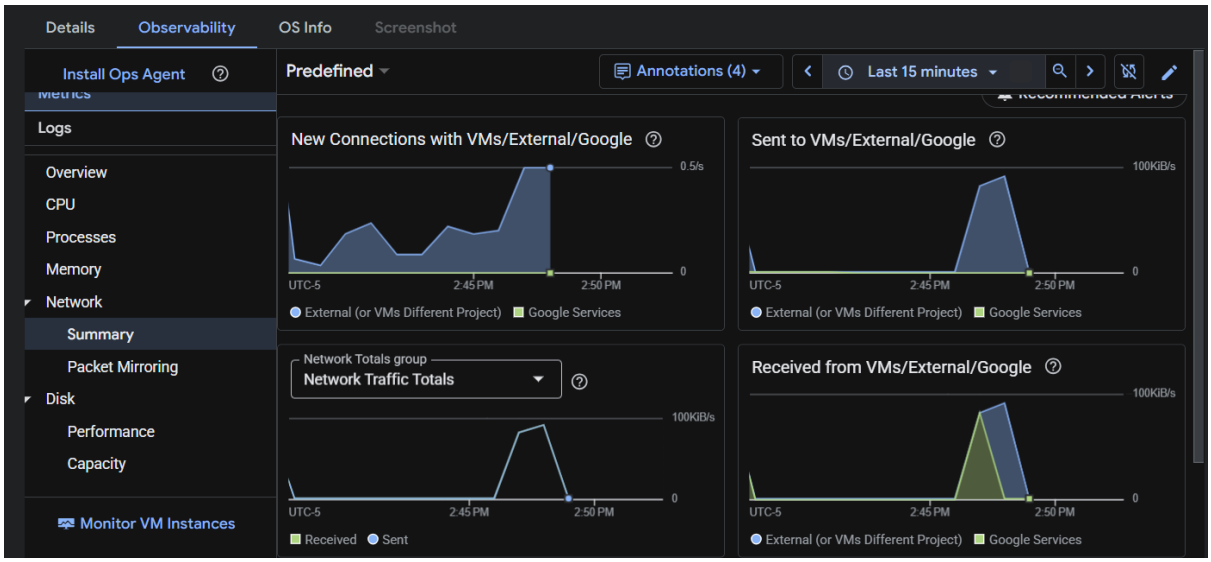
Question 4

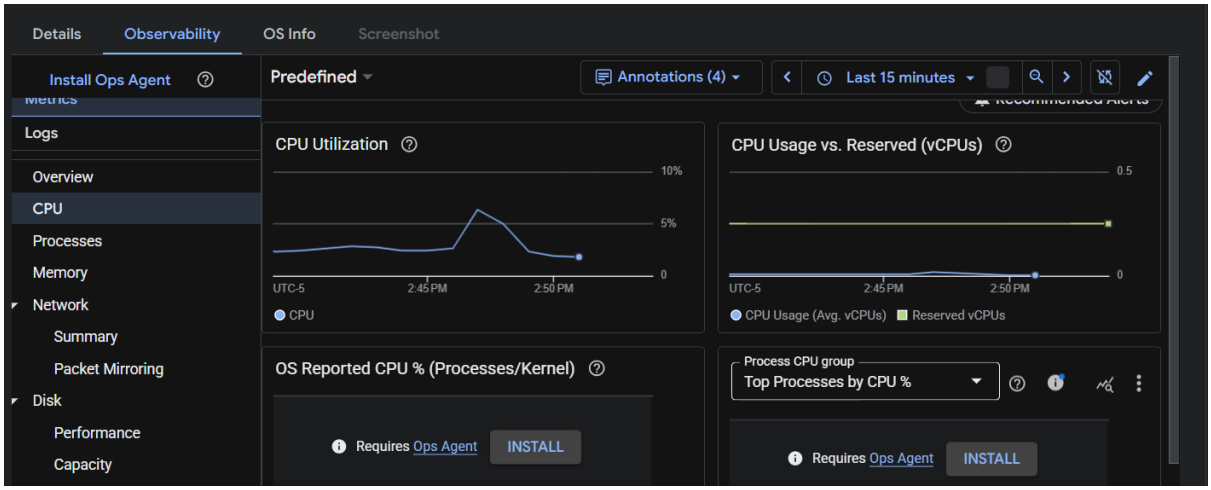
```
  [OK] GET /10047.html -> HTTP 200 (136928 bytes)
  [OK] GET /10048.html -> HTTP 200 (186575 bytes)
  [OK] GET /10049.html -> HTTP 200 (137942 bytes)
  [OK] GET /1005.html -> HTTP 200 (125407 bytes)
  [OK] GET /10050.html -> HTTP 200 (132434 bytes)
  [OK] GET /10051.html -> HTTP 200 (101836 bytes)
  [OK] GET /10052.html -> HTTP 200 (161024 bytes)
  [OK] GET /10053.html -> HTTP 200 (140959 bytes)
  [OK] GET /10054.html -> HTTP 200 (163011 bytes)
  [OK] GET /10055.html -> HTTP 200 (79278 bytes)
  [OK] GET /10056.html -> HTTP 200 (181062 bytes)
  [OK] GET /10057.html -> HTTP 200 (114877 bytes)
  [OK] GET /10058.html -> HTTP 200 (69245 bytes)
  [OK] GET /10059.html -> HTTP 200 (124400 bytes)
  [OK] GET /1006.html -> HTTP 200 (172535 bytes)
  [OK] GET /10060.html -> HTTP 200 (124938 bytes)
  [OK] GET /10061.html -> HTTP 200 (116376 bytes)
  [OK] GET /10062.html -> HTTP 200 (109870 bytes)
  [OK] GET /10063.html -> HTTP 200 (22614 bytes)
  [OK] GET /10064.html -> HTTP 200 (61228 bytes)
  [OK] GET /10065.html -> HTTP 200 (132442 bytes)
  [OK] GET /10066.html -> HTTP 200 (92312 bytes)
  [OK] GET /10067.html -> HTTP 200 (29126 bytes)
  [OK] GET /10068.html -> HTTP 200 (105855 bytes)
  [OK] GET /10069.html -> HTTP 200 (16095 bytes)
  [OK] GET /1007.html -> HTTP 200 (110376 bytes)
  [OK] GET /10070.html -> HTTP 200 (6068 bytes)
  [OK] GET /10071.html -> HTTP 200 (17097 bytes)
  [OK] GET /10072.html -> HTTP 200 (118898 bytes)
  [OK] GET /10073.html -> HTTP 200 (170542 bytes)
  [OK] GET /10074.html -> HTTP 200 (129917 bytes)
  [OK] GET /10075.html -> HTTP 200 (113870 bytes)
  [OK] GET /10076.html -> HTTP 200 (185574 bytes)
  [OK] GET /10077.html -> HTTP 200 (133938 bytes)
  [OK] GET /10078.html -> HTTP 200 (87311 bytes)
  [OK] GET /10079.html -> HTTP 200 (91302 bytes)
  [OK] GET /1008.html -> HTTP 200 (68749 bytes)
  [OK] GET /10080.html -> HTTP 200 (123918 bytes)
  [OK] GET /10081.html -> HTTP 200 (71758 bytes)
  [OK] GET /10082.html -> HTTP 200 (134419 bytes)
  [OK] GET /10083.html -> HTTP 200 (160013 bytes)
  [OK] GET /10084.html -> HTTP 200 (125915 bytes)
  [OK] GET /10085.html -> HTTP 200 (28635 bytes)
  [OK] GET /10086.html -> HTTP 200 (94336 bytes)

[Client] Done: 100 success, 0 failed in 11.30s
[Client] Throughput: 8.9 req/s
Aum@hw4-client-vm:~$
```

Logs and Monitoring

1) bash /opt/client/run_client.sh

```
Aum@hw4-client-vm:~$ bash /opt/client/run_client.sh
[Client] Target server: 136.113.118.31:80
[Client] Requesting 100 files ...
  [OK] GET /0.html -> HTTP 200 (99334 bytes)
  [OK] GET /1.html -> HTTP 200 (117885 bytes)
  [OK] GET /10.html -> HTTP 200 (179070 bytes)
  [OK] GET /100.html -> HTTP 200 (96825 bytes)
  [OK] GET /1000.html -> HTTP 200 (173546 bytes)
  [OK] GET /10000.html -> HTTP 200 (51190 bytes)
  [OK] GET /10001.html -> HTTP 200 (33146 bytes)
  [OK] GET /10002.html -> HTTP 200 (41672 bytes)
  [OK] GET /10003.html -> HTTP 200 (43177 bytes)
  [OK] GET /10004.html -> HTTP 200 (117384 bytes)
  [OK] GET /10005.html -> HTTP 200 (2559 bytes)
  [OK] GET /10006.html -> HTTP 200 (34155 bytes)
```





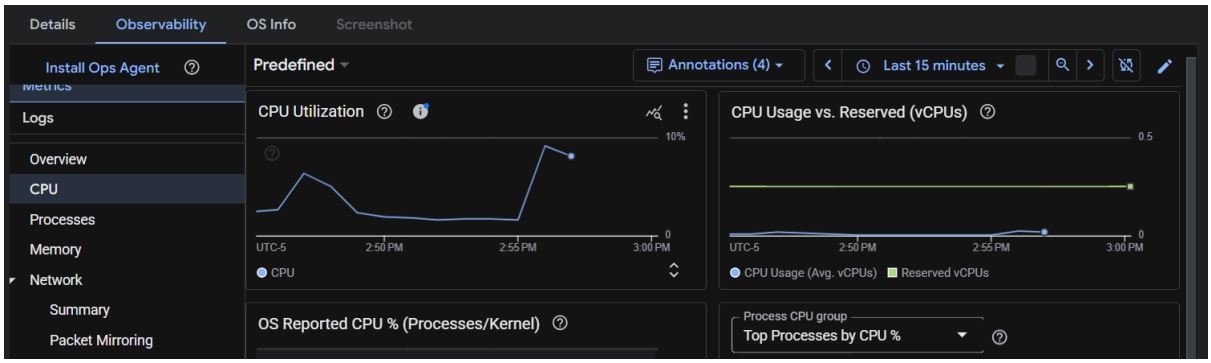
2) bash /opt/client/run_client.sh & bash /opt/client/run_client.sh & wait

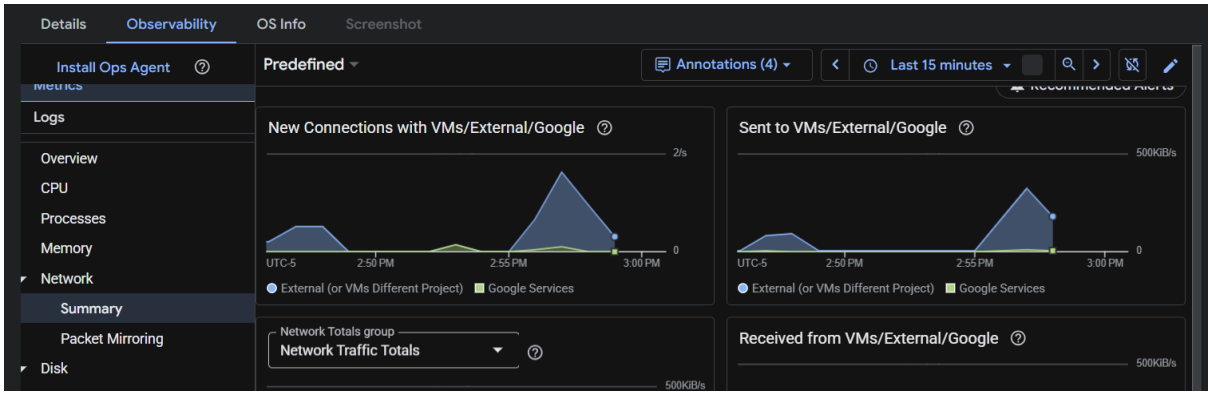
Original Memory in idle

```
Aum@hw4-client-vm: ~
Every 2.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         389         453           0         279         580
Swap:             0           0           0
```

When running the script

```
Aum@hw4-client-vm: ~
Every 2.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         454         387           0         279         515
Swap:             0           0           0
```





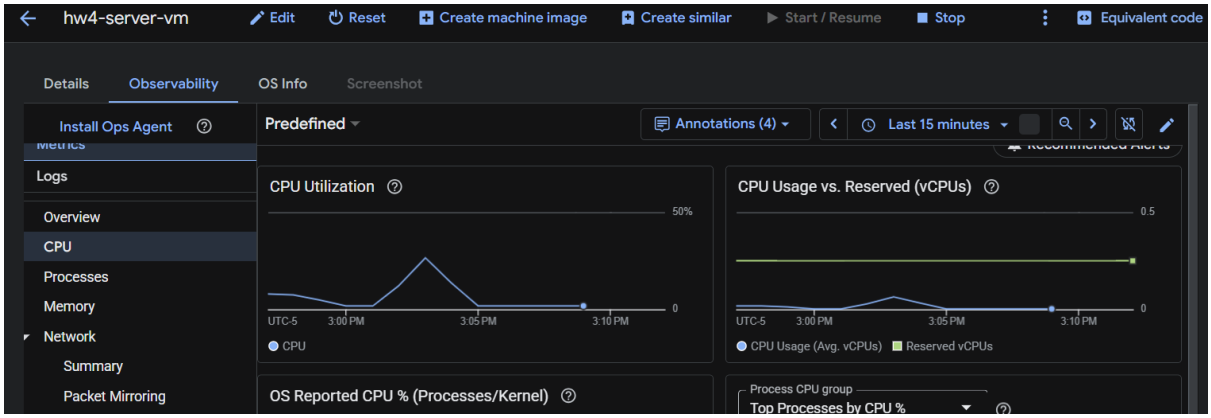
3) for i in 1 2 3; do bash /opt/client/run_client.sh & done; wait

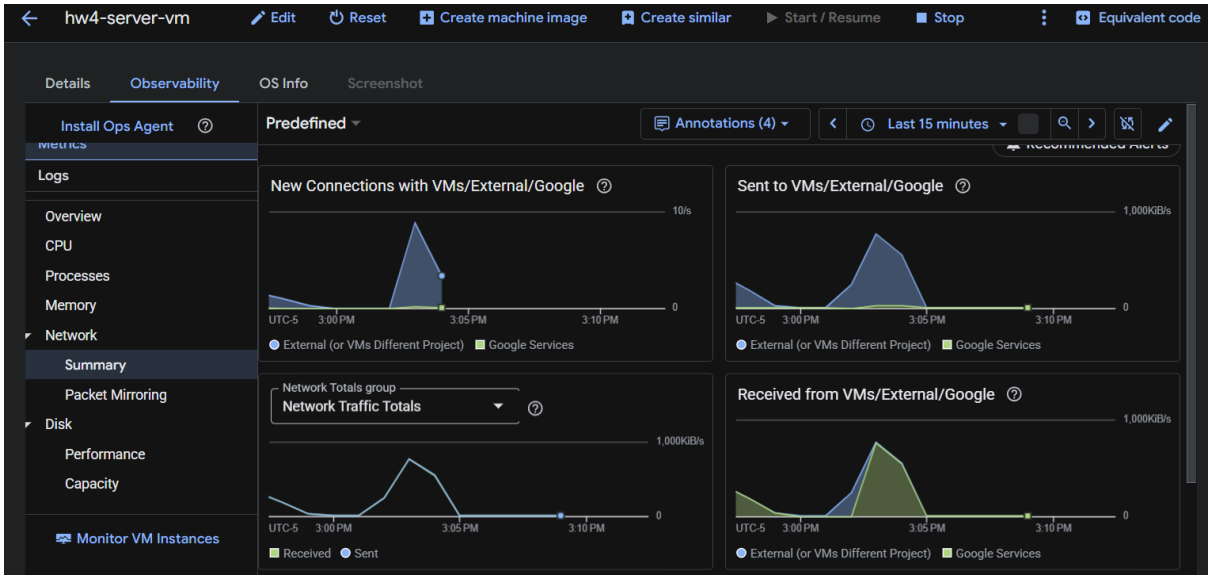
Original memory in idle

```
Aum@hw4-client-vm: ~
Every 5.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         390         452           0         279         579
Swap:             0           0           0
```

After running the script

```
Aum@hw4-client-vm: ~
Every 5.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         489         353           0         279         480
Swap:             0           0           0
```





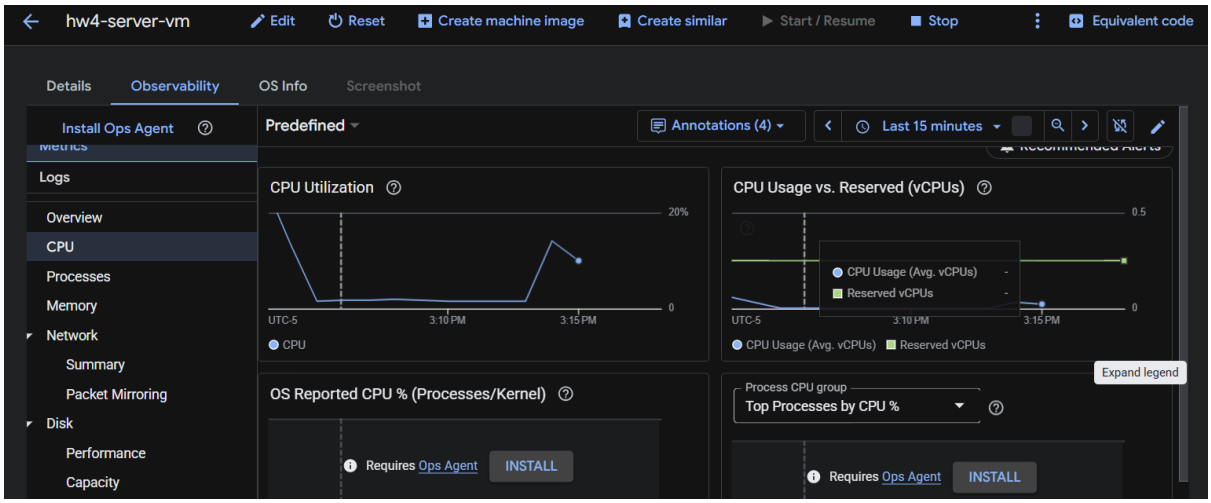
4) for i in 1 2 3 4; do bash /opt/client/run_client.sh & done; wait

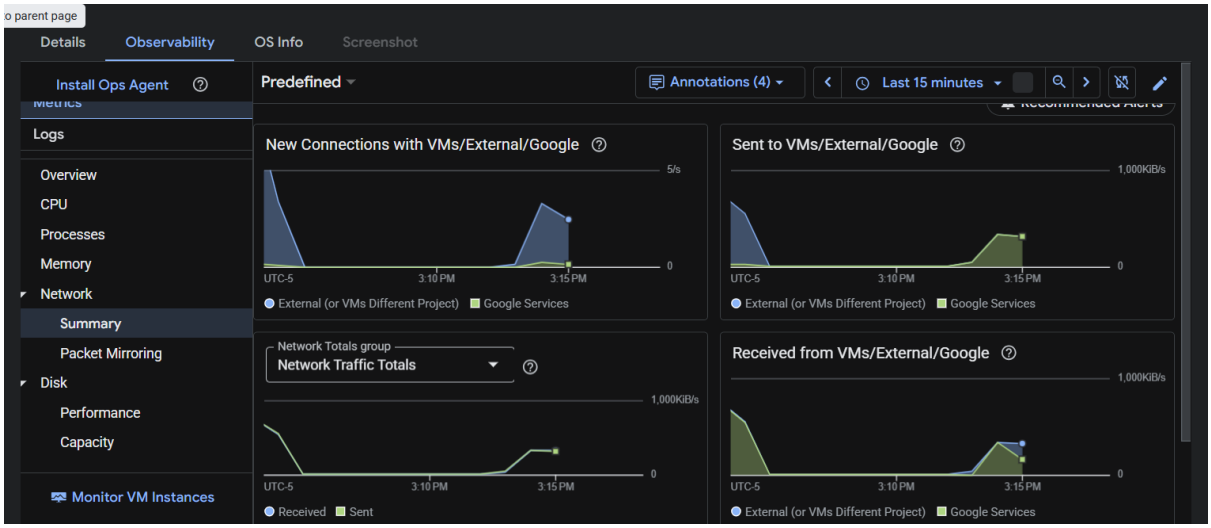
Original memory in idle

```
Aum@hw4-client-vm: ~
Every 5.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         388         517           0         214         581
Swap:             0           0           0
```

After running the script

```
Aum@hw4-client-vm: ~
Every 5.0s: free -m

              total        used        free      shared  buff/cache   available
Mem:            969         516         389           0         214         453
Swap:             0           0           0
```





Performance Table

Clients	Total Requests	Errors	Avg Throughput/client	Total Throughput	Avg Time
1	100	0	10.5 req/s	10.5 req/s	9.52s
2	200	0	13.9 req/s	27.8 req/s	7.19s
3	300	0	14.7 req/s	44.0 req/s	6.83s
4	400	0	16.6 req/s	66.2 req/	66.2s

Findings from stress testing hw4-server-vm (e2-micro, 2 vCPU, 1 GB RAM):

The web server did not return any errors across all 4 levels of concurrent clients. Total throughput scaled from 10.5 req/s with 1 client to 66.2 req/s with 4 clients - a 6.3x improvement.

Why throughput increased (not decreased) with more clients: The bottleneck in this workload is not the VM's CPU or RAM - it is network I/O to Google Cloud Storage. Each request spawns a thread that blocks on a GCS download. With more concurrent clients, the server overlaps multiple GCS fetch operations simultaneously. This means the GCS bandwidth is utilized more efficiently, reducing per-request wait time.

CPU remained low because each thread is mostly idle, waiting for GCS network responses - not doing compute work.

Memory increased slightly with more threads, but stayed well within e2-micro's 1 GB RAM limit because the ThreadedHTTPServer design streams responses without buffering all 100 files at once.

Bandwidth scaled proportionally with concurrent clients - 4x clients ≈ 4x data transferred simultaneously.

Peak resource usage over 3 minutes (from GCP Monitoring): [fill in from your monitoring screenshots - CPU %, memory %, network bytes/s].

The server could not be saturated with 4 clients on this workload because GCS scales horizontally. A workload that was CPU-intensive (e.g., image processing) would have hit e2-micro limits much sooner.

Question 9 - Github Repo - https://github.com/yourname/AUM-U07079548-CLOUD-COMPUTING-HW/tree/main/hw4